

Volatility Spillover Effects and Conditional Correlations in Bangladesh and Indian Stock Markets

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Abstract

This paper examines the volatility dynamics of stock markets between Bangladesh and India over the period from 28 November 2001 to 20 February 2017. Results of a bivariate VAR model reveal the strong evidence of unidirectional spillovers in returns from the Bombay Stock Exchange (BSE) of India to Dhaka Stock Exchange (DSE) of Bangladesh. Applying preliminary statistics and univariate EGARCH models, we find that the return series of DSE and BSE show evidence of leptokurtosis, volatility clustering and leverage effect. The volatility transmissions across markets are examined using augmented GARCH(1,1) model. Results explore that the volatility spillover effects are bidirectional between BSE and DSE, while India is the largest volatility transmitter both domestically and internationally. Finally, we find evidence of very low conditional correlation between BSE and DSE though the estimated conditional correlation is positive and significantly different from zero at better than the 5% significance level.

Key words: Stock Market, Volatility Spillover, VAR, EGARCH, MGARCH-CCC

JEL classification: C32, C58, F36, G15

1. Introduction

The noteworthy rise in global financial flows by the increasing globalized economic activity lead to interdependence of major financial markets all over the world (Balios and Xanthakis, 2003). The existence of robust trading and economic links, the escalation in liberalization activities of governments, the expansion of international finance and trade, swift developments in trading systems and telecommunication, and the establishment of common trading blocs, such as, European Union, NAFTA, SAARC and ASEAN are some factors contributing to financial integration (Aktan et al. 2009). Although financial globalization, integration, co-movement and volatility spillover can carry some financial crisis, such as, Asian financial crisis in 1997-98 and global financial crisis in 2007-08, it generate most important economic benefits as it assist investors to diversify risks worldwide. Harris and Pisedtasalasai (2006) argue that volatility spillover is an indicator of market efficiency. Thus, the analysis of financial market co-movement and volatility spillover between stock returns has been a central research agenda in economics and finance over the few years. Co-movement between stock markets refers to a phenomenon of a return series moving with another return series. Volatility spillover or contagion refers to the spread of market disturbances from one market to another,

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a process observed through comovements in stock prices, exchange rates or capital flows (Dornbusch et al. 2000). It is evident from the literature that not only the developed countries but also the financial markets of developing countries become interrelated (Aktar, 2009). So, a proper attention should be given to check the co-movement and volatility spillover of the stock markets between Bangladesh and India, as Beine and Candelon (2007) argue that the presence of strong policy coordination and economic ties between related countries can connect their stock indices over time.

Due to globalization and financial liberalization, Bangladesh is also in a process of interaction and integration with rest of the world. Both Bangladesh and India moved into the process of financial liberalization since early 1990s. Bangladesh is surrounded in three sides by India and have a sound knowledge over each other with a strong trade and economic ties. Foreign trade has been an important dimension in Indo-Bangla relations since the independence of Bangladesh in 1971. India was the largest trading partner of Bangladesh since 1971 until 2014 (Kashem, 2016). India's trade with Bangladesh has witnessed rapid growth in recent years. In terms of merchandise trade, the total trade has increased from USD 1 billion in FY2001 to around USD 4.4 billion in FY2012, while India's share in the bilateral trade is around 86 percent of the total trade (Acharya and Marwaha, 2012). Currently, total bilateral formal trade volume increases to about USD 6.5 billion; however, India's annual earnings from formal and informal trade with Bangladesh is somewhere near 14 to 15 billion dollars (Balaji, 2016). Bangladesh is recently positioned in top ten export destinations of India. Despite being the second largest source for imports, India is the only Asian country to emerge as one of the top ten export destinations of Bangladesh. With the recently signed twenty two important bilateral documents which together covers varied areas of cooperation, economics, trade and investment, security, infrastructure developments and education, India and Bangladesh are on the way of evolving a new paradigm for growing together harmoniously that could well become the model for other countries to follow (Balaji, 2016).

Thus, the increasing trade relations and the financial flows between Bangladesh and India have made us to expect a co-movement among the stock markets of the two nations. According to Dekker, Sen and Young (2001), markets with strong economic ties and close geographic proximity are more closely linked than the isolated market. Morelli (2010) and Febrian and Herwany (2007) also give attention to the markets with close geographic proximity. Moreover, Hasan and Wadud (2015) reveal that Indian stock markets have some significant influence on Bangladeshi stock markets in the short and long run. In this line of thinking, this study takes an attempt to examine the possible inter-linkages between stock markets in Bangladesh and India. Trade links, regional patterns and macroeconomic similarities make countries vulnerable to volatility (Dornbusch et al. 2000). Considering the facts, this study also takes an attempt to investigate volatility transmission across stock returns of Bangladesh and India. Specifically, this study focuses on following objectives:

- a. To study the possibility of spillover in stock returns over time in Bangladesh and India through use of VAR-Granger Causality model;
- b. To capture the nature of stock returns volatility applying univariate EGARCH model;
- c. To detect transmission of volatility between stock markets using Augmented GARCH model;
- d. To investigate the conditional correlation between stock markets in Bangladesh and India employing MGARCH-CCC framework.

This paper should contribute to the literature because hardly an effort has been given on co-movement and volatility spillover between stock returns in Bangladesh and India. Thus, this study will be useful for both investors and regulators in decision making. The study is organized in six sections as follows: section 2 presents the literature review; section 3 reports data and preliminary statistics, while section 4 justifies the methodology. Finally, section 5 reports the empirical findings and section 6 concludes the study.

2. Literature Review

This study is mostly motivated by several reasons. Most studies examine the co-movement and volatility spillover effects across stock markets focusing mainly on developed nations, such as, the U.S., Japan, and Europe. A little attention is also paid to emerging markets of Europe, Latin-America and Asia. Very few studies about linkages among stock markets have focused on South Asian region. South Asian markets, specifically, Bangladesh and India have experienced outstandingly rapid economic growth in the past one and half decades and are getting rising influence in the world capital markets. Thus, the linkages among these two emerging markets deserve closer attention. Moreover, Li and Majerowska (2008) argue that volatility should also be investigated when studying interactions among stock markets. But, studies on volatility spillover effects are scarce in emerging markets in general and in the South Asian region in particular. Thus, this study offers policymakers the opportunity to learn about the level of stock market correlation and volatility spillover outcomes between Bangladesh and India. This investigation also offers foreign investors the opportunity to learn about the extent of diversification benefits which this two country can offer.

Dedi and Yavas (2016) investigate linkages among equity market returns and volatility spillovers in Germany, United Kingdom, China, Russia, and Turkey applying MARMA, GARCH, GARCH-in-mean, and EGARCH methodologies to daily data on country exchange-traded funds based on the MSCI indices from 31 March 2011 to 11 March 2016. The results of the analysis show the existence of significant co-movements of returns among the countries in the sample. They also reveal that all of the countries, with the exception of UK and Turkey, experience volatility spillovers from other markets. **Singh and Kaur (2016)** investigate the co-movement in the BRIC countries' stock markets in the long run employing a Johansen cointegration technique. The results indicate no long run co-movement among the BRIC countries as a whole. However, the pairwise and multivariate cointegration tests highlight the existence of a co-movement among the Brazilian, Russian and the Chinese markets, excluding Indian during the financial crisis and the period afterwards. **Mohammadi and Tan (2015)** examines the dynamics of daily returns and volatility in stock markets of the U.S., Hong Kong and mainland China (Shanghai and Shenzhen) over 2 January 2001 to 8 February 2013 employing VAR and MGARCH models. The results suggest the evidence of unidirectional return spillovers from the U.S. to the other three markets; but no spillover between Hong Kong and either of the two mainland China markets. The study also finds the evidence of unidirectional ARCH and GARCH effects from the U.S. to the other three markets. The patterns of dynamic conditional correlations from the DCC model suggest an increase in correlation between China and other stock markets since the most recent financial crisis of 2007. **Gahlot (2013)** intends to study volatility spillover among South Asian Countries (Bangladesh, India, Pakistan and Sri Lanka) and U.S. stock markets through use of Granger causality test and C GARCH M. He finds significant bidirectional causality between Stock market of U.S. and India for both short terms as well as for long term which is not disturbed by recession. But the recession has changed causal relation among other countries. The study also reveals that volatility of all South Asian countries is of long term nature and the observed

spillover effects are unstable over time in sense that the spillover changed its nature after beginning of recession. **Padhi and Lagesh (2012)** investigate volatility transmission using daily stock-price indices from July 1, 1994, through September 30, 2009 for five Asia equity markets, India and United States applying bivariate BEKK and DCC model for the analysis. They find that bilateral shocks and volatility spill over exists in the India/Malaysia, India/Taiwan and India/Indonesia. Evidence demonstrates that Indonesia is the main transmitter within the Asia markets. The significance of DCC-GARCH estimates explains that conditional correlation between India and the Asia-US stock markets are highly dynamic and time varying. **Ali, Butt and Rehman (2011)** examine the co movement of Pakistan's Equity Market with the markets of India, China, Indonesia, Singapore, Taiwan, Malaysia, Japan, USA and UK by using co integration test on monthly stock prices from the period of July 1998 to June 2008. The results reveal that there is no co movement of Pakistan's equity market with the markets of UK, USA, Taiwan, Malaysia and Singapore. Whereas the stock prices of Pakistan equity market move together with the stock prices of India, China, Japan and Indonesia so there is no chance of risk minimization for investors through international portfolio in these countries. **Abou-Zaid (2011)** investigates the international transmission of daily stock index volatility movements from U.S. and U.K. to selected MENA emerging markets: Egypt, Israel, and Turkey employing a multivariate GARCH in Mean technique for the period from January 2, 1997 to September 25, 2007. The study finds that Egypt and Israel are significantly influenced by the U.S. stock market while Turkey is not. **Hoque (2007)** explores the dynamics of stock price movements of Bangladesh with USA, Japan and India employing Johansen and Juselius multivariate cointegration approach, vector error correction model and impulse response analysis. He finds an evidence of cointegration among the markets demonstrating that stock prices in the countries share a common stochastic trend. Impulse response analysis shows that shocks to the US market do have an impact on the Bangladesh market. The evidence of Bangladesh stock market responding to shocks in the Indian market is weak. Finally, the study reveals that shocks to the Japanese market do not generate a response in the Bangladesh market. **Savva, Osborn and Gill (2005)** investigate the transmission mechanism of price and volatility spillovers across the New York, London, Frankfurt and Paris stock markets under the framework of the multivariate EGARCH model. Also, the correlation between those markets is investigated for the periods before and after the introduction of euro under the Constant and Dynamic Conditional Correlation frameworks. By using daily closing prices, they find evidence that the domestic stock prices and volatilities are influenced by the behavior of foreign markets. The volatility is found to respond asymmetrically to news in other markets. The findings also indicate that the correlations of returns have increased for all markets since the launch of euro, with that between Frankfurt and Paris experiencing the largest increase. **McAleer and Veiga (2005)** finds that the New York and Tokyo markets experience positive and significant spillovers from the other market. **Bekaret and Wu (2000)** investigate volatility spillover of stock market returns in the U.S. and Japan. The study finds evidence of direct volatility spillover running mainly from U.S. stock market to Japan's stock market. **Bekaert and Harvey (1997)** analyse the volatilities of emerging equity markets and found that the volatility is strongly influenced by global factors in the fully integrated markets but is more likely to be influenced by local factors in the segmented market. **Liu and Pan (1997)** investigate stock return and volatility spillover effects from the U.S. and Japanese markets to Hong Kong, Singapore, Taiwan, and Thailand stock markets. The result of their study reveal that the US market is more influential than the Japanese market in transmitting return and volatility to the Asian market. Furthermore, the evidence of their study indicates that while the cross-country stock investing hypothesis cannot by itself explain the international transmissions of return and volatility, the market contagion also plays an important role in the transmission mechanism.

3. Data and Preliminary Statistics

The empirical work in this study requires daily data on stock prices for the Dhaka Stock Exchange (DSE) of Bangladesh and the Bombay Stock Exchange (BSE) of India. Dhaka Stock Exchange is the oldest and largest stock exchange in Bangladesh. The DSE is very young as its extensive activity started from 1990s though it was formally started its trading in 1956. Bangladesh has turned into an independent sovereign state on 16 December 1971. Trading activity of the DSE suspended for the start of the liberation war until it restarted in 1976. When restarted in 1976, DSE had only 9 listed companies with a paid-up capital approximately 137.52 million Taka. As on February 20, 2017, the market capitalization of 563 listed securities of DSE is 37,56,668 million Taka. DSE used three share price indices- All Share Price Index (DSI), General Price Index (DGEN) and DSE-20 before starting Standards & Poor's supported DSE Broad Index (DSEX) and DSE-30 Index in 28 January 2013. DGEN was considered as the benchmark index of DSE before starting DSEX calculation. Thus, this study purposively uses both DGEN (up to 27 January 2013) and DSEX indexes (from 28 January 2013). Bombay Stock Exchange (BSE) was established in 1875 and it is the oldest stock exchange in the entire Asia. With more than 5500 listed companies, the BSE is the world's 11th largest stock exchange with an overall market capitalization of 1.43 Trillion U.S. Dollar as of March 2016. The S&P SENSEX (BSE-30) is the main benchmark index of BSE. This study uses close-of-the-day index prices cover the period from 28 November 2001 to 20 February 2017. Due to holidays, the stock markets in Bangladesh and India might be closed on different days. To address this, following Li (2007), we omit all observations with missing values. Following this adjustment, there are a total of 2830 daily observations. The daily index data of two indices obtained from DSE and BSE websites are used to calculate returns as follows:

$$R_{i,t} = [\text{Log}(P_{i,t}) - \text{Log}(P_{i,t-1})] \times 100$$

where, R_i = Daily return, Log = Natural Logarithms, P_t = Price Index at time t , and P_{t-1} = Price Index at time $t-1$.

Figure 1 presents the daily rates of return for the two adjusted indexes. Daily returns fluctuate around zero and are characterized by volatility clustering. DSE returns demonstrate higher volatility in 2010–2011, while BSE returns exhibit higher volatility in 2007–2009.

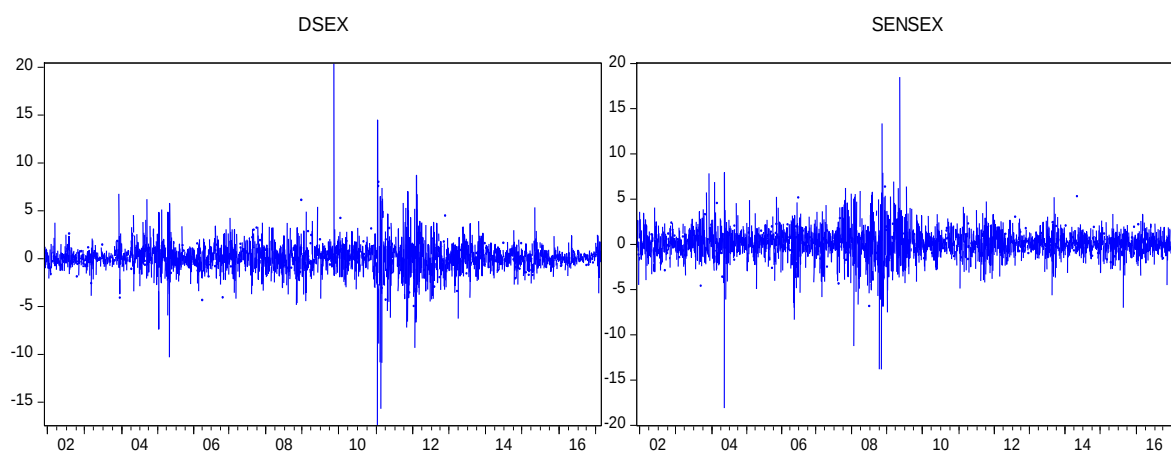


Figure 1. Daily rates of return in the two stock indices.

Descriptive statistics of the daily returns of DSE and BSE are shown in Table 1. It is evident from the table that average daily returns are positive across the two markets. The daily returns of BSE are larger than the daily returns of DSE. DSE is the less volatile series though the two

markets exhibit almost similar degrees of volatility as reflected in their standard deviations. Both series have small negative skewness. The negative skewness in DSE and BSE implies that large negative changes in returns occur more often than positive changes. We observe that the kurtosis of 25.54 and 17.21 of DSE and BSE respectively imply that distributions are leptokurtic, i.e., both return series have fatter tails than a normal distribution where there are higher likelihood of large gains or large losses on an investment. This excess kurtosis also indicate that the volatility of the investment in DSE and BSE is high. It suggests that stock market returns of Bangladesh and India exhibit leptokurtosis that is a well-known stylized fact in the finance literature. The large Jarque-Bera statistics reject the null hypothesis of normal distribution for both series. Therefore, the rejection of the normality test based on Jarque-Bera test gives evidence for the existence of GARCH effects. Finally, the volatility clustering nature of DSE and BSE returns is confirmed by the autocorrelation test that is also reported in Table 1. The Ljung -Box Q and Q^2 statistics show that there are significant serial correlations for both stock market returns. This may be seen as evidence for the presence of volatility clustering in both the returns series. These results also indicate that the return series of both markets exhibits conditional heteroskedasticity and that a GARCH process is a likable candidate for modeling their time-series behavior.

Table 1. Basic Statistics

	Mean	Median	Max.	Min.	Std. Dev.	Skewness	Kurtosis	Jarque-Bera	Q (36)	Q ² (36)	Obs.
DSE	0.07	0.08	20.39	-17.4	1.63	-0.09	25.54	59917.79 (0.00)	88.48 (0.00)	880.73 (0.00)	2829
BSE	0.08	0.10	18.49	-18.1	1.75	-0.31	17.21	23838.52 (0.00)	79.24 (0.00)	837.23 (0.00)	2829

The results of Augmented Dickey Fuller (ADF) and Phillips-Peron (PP) unit root tests in Table 2 reveal that the null hypothesis of unit root is strongly rejected at one percent significant level for the DSE and BSE returns. It confirms that all the return series are stationary, that is, they do not follow a random walk. Since, both the return series are stationary, we can follow GARCH processes.

Table 2. ADF and PP Unit Root Test Results of the Variables

Variables	ADF			PP		
	Intercept	Trend & Intercept	None	Intercept	Trend & Intercept	None
DSE	-10.20* (0.00)	-10.23* (0.00)	-10.03* (0.00)	-54.25* (0.00)	-54.25* (0.00)	-54.24* (0.00)
BSE	-14.49* (0.00)	-14.53* (0.00)	-14.33* (0.00)	-52.95* (0.00)	-52.97* (0.00)	-52.85* (0.00)

Note: First bracket shows P-values. * indicates stationary at 1% significant level using MacKinnon (1996) critical and P -values.

4. Models

This study examines the possibility of spillover in stock returns in Bangladesh and India using VAR-Granger Causality model. In order to get reliable results of volatility dynamics, researchers should initially examine the volatility characteristics of stock returns, such as, heavy tails, volatility clustering and leverage effects. Miron and Tudor (2010) argue that stock returns exhibit some patterns and that are crucial for correct model specification and estimation.

In this line of thinking, we have already checked that the return series of DSE and BSE show evidence of volatility clustering and leptokurtosis. The stationary properties of the return series have been also checked applying ADF and PP unit root tests. Further, we intend to see whether these return series follow the asymmetry or leverage effect employing EGARCH model. Then, we try to investigate the stock returns volatility spillover effect between Bangladesh and India employing Augmented GARCH model. Finally, we employ MGARCH-CCC framework to examine the correlation between stock markets in Bangladesh and India.

4.1 Vector Autoregressive (VAR) Model

Vector Autoregression (VAR) is used to find the relationship between various variables by taking into account the feedback by other variables. Estimation based on VAR contains endogenous variables in adding exogenous variables in the equation, making it an important tool for multivariate analysis. As in Enders (2004), an n -equation VAR can be represented by:

$$\begin{bmatrix} X_{1t} \\ X_{2t} \\ \vdots \\ X_{nt} \end{bmatrix} = \begin{bmatrix} A_{10} \\ A_{20} \\ \vdots \\ A_{n0} \end{bmatrix} + \begin{bmatrix} A_{11}(L) & A_{12}(L) & \dots & A_{1n}(L) \\ A_{21}(L) & A_{22}(L) & \dots & A_{2n}(L) \\ \vdots & \vdots & \ddots & \vdots \\ A_{n1}(L) & A_{n2}(L) & \dots & A_{nn}(L) \end{bmatrix} \begin{bmatrix} X_{1t-1} \\ X_{2t-1} \\ \vdots \\ X_{nt-1} \end{bmatrix} + \begin{bmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \\ \vdots \\ \varepsilon_{nt} \end{bmatrix} \quad (1)$$

where A_{i0} = the parameters representing intercept terms, $A_{ij}(L)$ = the polynomials in the lag operator L . The individual coefficients of $A_{ij}(L)$ are denoted by $a_{ij}(1)$, $a_{ij}(2)$,... Since all equations have the same lag length, all the polynomials $A_{ij}(L)$ are of the same degree. The terms ε_{it} are white-noise disturbances that may be correlated. Again designate the variance/covariance matrix by Σ , where the dimension of Σ is $(n \times n)$.

This study examines the possibility of spillovers in returns over time and across market using a two-variable VAR model consisting of returns for DSE and BSE. A two-equation VAR can be represented by where the returns of market 1 depend on p lags of itself, as well as p lags of the returns of market 2:

$$R_{1t} = \varphi_1 + \sum_{i=1}^p \varphi_{1i} R_{1,t-i} + \sum_{i=1}^p \varphi_{2i} R_{2,t-i} + \varepsilon_{1,t} \quad (2)$$

The possibility of spillover in returns over time can be examined by testing the joint hypotheses that $\varphi_{1i} = 0$ ($i = 1, \dots, p$). Similarly, the possibility of spillovers in returns from Market 2 to Market 1 can be examined by testing for the joint hypotheses that $\varphi_{2i} = 0$ ($i = 1, \dots, p$).

4.2 EGARCH Model

This study uses Exponential Generalized Autoregressive Conditional Heteroskedasticity (EGARCH) model extended by Nelson (1991) from Autoregressive Conditional Heteroskedasticity (ARCH) model of Engle (1982) and Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model of Bollerslev (1986). The conditional variance equation of EGARCH (1, 1) model can be written as:

$$\ln \sigma_t^2 = \omega + \alpha |z_{t-1}| + \gamma z_{t-1} + \beta \ln \sigma_{t-1}^2 \quad (3)$$

where the left hand side is the logarithm of the conditional variance. It implies that the leverage effect is exponential and that forecasts of the conditional variance are guaranteed to be non-negative. z_{t-1} shows the asymmetric impact of positive and negative shocks. The asymmetry term $\gamma < 0$ implies that negative shocks have a greater impact on volatility rather than the positive shocks. The negative asymmetric term also suggests for leverage effect that negative shocks do obviously have a bigger impact on future volatility than positive shocks of the same magnitude.

4.3 Augmented GARCH Model

In order to capture the co-movement volatility between Bangladesh and Indian stock markets, this study uses augmented GARCH model developed by Duan (1997) as it is considered to be very versatile univariate volatility model (Dedi and Yavas, 2016). Moreover, Edwards (1998), Zouch et al. (2011) and Dedi and Yavas (2016) use this method and detect the presence of volatility transmission effect.

The Augmented GARCH model as developed by Duan (1997) can be written as:

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 + \theta X_t \quad (4)$$

where X_t is the residual squared of ARMA model and θ is the term that measures the magnitude of volatility transmission across the markets.

As in Dedi and Yavas (2016), this study employs one-period lagged squared DSE and BSE returns for X_t . Therefore, we report transmission of volatility of past period returns as opposed to transmission of conditional volatilities.

4.4 MGARCH-CCC

Constant Conditional Correlation (CCC) model is developed by Bollerslev (1990). This model assumes that correlations between each pair of returns are constant and thus the volatility model consists only of the equations for the variances. CCC model has been very popular among empirical studies because it reduces the conditional correlation matrix to constant correlation coefficients between variables, so the number of parameters to be estimated is small in comparison with other models. The conditional covariance matrix is defined as:

$$H_t = S_t R S_t \quad (5)$$

where S_t is a $(N \times N)$ diagonal matrix of time-varying standard deviations and R is a $(N \times N)$ matrix of constant correlations.

In the bivariate case as in this study, the CCC becomes as follows:

$$H_t = \begin{bmatrix} \sqrt{h_{11,t}} & 0 \\ 0 & \sqrt{h_{22,t}} \end{bmatrix} \begin{bmatrix} 1 & \rho_{12} \\ \rho_{21} & 1 \end{bmatrix} \begin{bmatrix} \sqrt{h_{11,t}} & 0 \\ 0 & \sqrt{h_{22,t}} \end{bmatrix} \quad (6)$$

In this case, H_t is assumed to be positive definite if certain restrictions on the parameters are correctly satisfied. Variance terms $h_{11,t}$ and $h_{22,t}$ are univariate GARCH processes with $p=q=1$.

5. Results and Discussion

This section registers the results of estimating the VAR, EGARCH(1,1), Augmented GARCH(1,1) and MGARCH(1,1)-CCC models utilizing the returns in the two equity markets. We apply GARCH processes with $p=q=1$, as Alexander (2001) argues that it is rarely necessary to use more than a GARCH(1,1) model. All empirical work is carried out using the EViews 9 software package.

5.1 VAR-Granger Causality Results

In order to estimate the equation (2), an appropriate lag length must be determined. So, we use and check five different criteria namely, Likelihood Ratio (LR), Final Prediction Error (FPE), Akaike Information Criteria (AIC), Schwarz Information Criteria (SIC) and Hannan-Quinn Information Criteria (HQ) to determine the optimum lag lengths of the VAR model. Results for each criterion with a maximum of 24 lags exhibit that AIC, sequential modified LR and FPE criteria stand in favor of 18 lags, while SIC and HQ criteria suggest for only 0 lag. The presence of residual serial correlation makes the result less efficient. Thus, we proceed to conduct LM tests for each suggested lags up to maximum 24 lags. Using 18 lags suggested by LR, AIC and FPE criterion produce no autocorrelation in the VAR model for up to 24 days at 1% significance level. So, we accept VAR (18) model for Granger Causality/Block Exogeneity Wald tests between two markets.

The reported results in Table 3 reveal that there is evidence of unidirectional spillovers in returns from the Indian stock market to Bangladesh stock market, while there is no evidence of causality from DSE to BSE. It discloses strong impact of the Indian markets on Bangladesh stock market return movement which is consistence to the arguments that developed stock markets have significant influence in transmitting return and volatility to emerging stock markets (Liu and Pan, 1997).

Table 3. Estimates of VAR Granger Causality Test

Dependent	Excluded	Chi-sq	df	Prob.
DSE	BSE	41.61945*	18	0.0012
BSE	DSE	18.00573	18	0.4553

Note: * indicates significance at the 1% level.

5.2 EGARCH Results

Table 4 reports results of the variance equation of the estimated model for DSE returns and BSE returns. The estimated EGARCH(1,1) model show that all the parameters are highly significant at 1% level. This is a strong indication for a leverage effect in both DSE and BSE. In fact, the asymmetry terms γ i.e., $RESID(-1)/@SQRT(GARCH(-1))$ are negative and highly significant for DSE and BSE returns. This implies that negative shock has a greater impact on volatility rather than the positive shocks of the same magnitude in Dhaka and Bombay stock exchange. Thus, the volatility asymmetry indicates that investors of Bangladesh and India are more prone to the negative news in comparison to the positive news. In terms of diagnostic fit presented in Table 3, the estimated model satisfies all conditions of the GARCH theory based on Ljung -Box Q^2 statistics and ARCH-LM test.

Table 4. Estimates of the EGARCH(1,1) Model

	Parameter Estimates				Diagnostic Tests		
	ω	α	γ	β	Q (36)	Q ² (36)	LM (36)
DSE	-0.247* (0.000)	0.422* (0.000)	-0.111* (0.000)	0.926* (0.000)	121.13* (0.00)	3.99 (1.00)	4.13 (1.00)
BSE	-0.117* (0.000)	0.197* (0.000)	-0.091* (0.000)	0.973* (0.000)	39.93 (0.30)	33.37 (0.59)	34.84 (0.70)

Notes: P-values are in brackets. * means significant at 1% level. For the parameters, P-values are associated with Z-statistics, and for diagnostic fitting, P-values are associated with Ljung -Box Q and chi-square statistics.

5.3 Augmented GARCH Results

To detect transmission of volatility between the two stock markets, the results of Augmented GARCH(1,1) model are reported in Table 5. It is evident from the results that Bangladesh and Indian stock market volatilities are affected by its own past volatility. Moreover, the statistical significance of GARCH parameters is revealing the extent of volatility clustering. Most importantly, the coefficient of the both volatility spillover terms are positive, indicating that an increasing volatility in India increases volatility in the Bangladesh stock market and vice versa. However, lagged volatility persistence in Bombay stock exchange has a more positive effect on current volatility in Dhaka stock exchange over time. Finally, it is also important to note that the volatility transmissions are bidirectional but own spillovers are much larger than the cross-market spillovers.

Table 5. Estimates of the Augmented GARCH(1,1) Model

	Parameter Estimates				Diagnostic Tests		
	ω	α	β	θ	Q (36)	Q ² (36)	LM (36)
DSE	0.022* (0.000)	0.166* (0.000)	0.825* (0.000)	0.017* (0.000)	110.71* (0.00)	2.42 (1.00)	2.39 (1.00)
BSE	0.043* (0.000)	0.088* (0.000)	0.898* (0.000)	0.003* (0.002)	39.94 (0.29)	23.27 (0.95)	22.75 (0.96)

Notes: P-values are in brackets. * means significant at 1% level. For the parameters, P-values are associated with Z-statistics, and for diagnostic fitting, P-values are associated with Ljung -Box Q and chi-square statistics.

5.4 MGARCH-CCC Results

The performance of the MGARCH-CCC model is reported in Table 6. This model estimates the own ARCH and GARCH effect as well as the correlations between each of the two markets. The results suggest the existence of own ARCH and GARCH effects in both markets as all of the estimated parameters are significantly different from zero and significant at 1% level. The estimated conditional correlation is positive and significantly different from zero at better than the 5% significance level. The low conditional correlation (0.04) between Bangladesh and India reflect the presence of weak direct interconnections between their markets. Finally, according to the diagnostics for the CCC model, we fail to reject the null hypothesis of serial independence in the DSE and BSE up to 5 lags, but reject the null at 36 lags.

Table 6. Estimates of the MGARCH-CCC model

	Transformed Variance Coefficients			
	Coefficient	Std. Error	z-Statistic	Prob.
M(1)	0.122365*	0.011115	11.00901	0.0000
A1(1)	0.346734*	0.015200	22.81147	0.0000
B1(1)	0.689320*	0.012584	54.77622	0.0000
M(2)	0.048172*	0.007038	6.844532	0.0000
A1(2)	0.090160*	0.006086	14.81409	0.0000
B1(2)	0.898218*	0.007531	119.2672	0.0000
R(1,2)	0.035787**	0.015101	2.369882	0.0178
Diagnostic Tests (Portmanteau Tests for Autocorrelations)				
Lags	Q-Stat	Prob.	Adj Q-Stat	Prob.
1	1.499462	0.8267	1.499992	0.8266
5	29.49460	0.0785	29.53287	0.0778
36	261.35*	0.0000	263.0319	0.0000

Notes: * and ** mean significant at 1% and 5 levels respectively.

6. Conclusions

This study investigates whether there is stock market co-movement, volatility spillover and conditional correlations between Bangladesh and India using daily stock return indices from 28 November 2001 to 20 February 2017. Employing VAR representation approach based on Granger causality, we find evidence of unidirectional return spillover, which is from the BSE to DSE. Thus we can comment that information from Indian market has made impact on the Bangladesh market. The preliminary statistics reveal that the return series of DSE and BSE show evidence of volatility clustering and leptokurtosis. Then, the results of the EGARCH (1, 1) model indicate that markets are more sensitive to the bad news than to a good news. The Augmented GARCH model is built to capture the existence of volatility spillover between returns of the DSE and BSE stock indices. The results reveal significant own past shocks and past volatility persistence impact on the current return fluctuation of both markets. Moreover, we find the existence of significant and positive bidirectional volatility transmission between these two markets. But, lagged volatility persistence in BSE has a more positive effect on current volatility in DSE over time. Although we find evidence of volatility spillovers, it is also important to note that own spillovers are much larger than the cross-market spillovers. Finally, we employ MGARCH-CCC framework to examine the conditional correlation between stock markets in Bangladesh and India. The results suggest the existence of own ARCH and GARCH effects in both markets. The low conditional correlation between Bangladesh and India reflect the presence of weak direct interconnections between their markets.

Based on the above discussion, we conclude that the volatility spillover effects are bidirectional between BSE and DSE, while India is the largest volatility transmitter both domestically and internationally. This finding implies that there are diminishing opportunities for investors to diversify their portfolios. Hence, stakeholders on the investment activity, including government, should pay attention on the behavior of volatility transmission. Private as well as institutional investors should modify their investment strategy and asset allocation decisions accordingly to the spillover effects. Future researchers can include one strong Asian and U.S. market with India to examine the spillover effect and conditional correlations from which Bangladesh may be strongly affected.

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